voltage spike a relay coil produces when switched off.

How it works

Fig. 2 shows the circuit diagram of the toggle-to-pulse switch converter. It is built around the popular NE555 timer IC and provides a simple, practical method for converting a toggle-switch input into a momentary pulse output. It generates a short, clean output pulse whenever a rising edge is detected at the input.

The 2-pin connector CON1 serves as the input terminal and accepts a 5V DC signal from an external toggle switch or other circuit. A push-to-on switch S1 is also provided for manual testing of the circuit.

Rising-edge detection is achieved through AC coupling using 47nF capacitor C5, which couples the input transition to the threshold pin6 of NE555 timer IC1. The threshold pin6 is normally held low by 10kΩ pull-down resistor R2, ensuring stable operation. A 1N4148 diode (D1) is connected across the input to suppress the response to the falling edge, allowing the circuit to respond primarily to the rising edge.

The RC network R4-C4 sets the pulse duration. When triggered, the NE555 generates a short pulse to energise relay RL1. To protect the circuit from back-EMF, a 1N4007 diode (D2) is wired across the relay coil, functioning as a freewheeling diode. A red LED (LED1) provides visual feedback, indicating the relay's status during operation. Thus, each rising edge at CON1 converts the toggle-switch signal into a momentary relay action. CON2 provides an isolated pulse output for interfacing with other circuits or loads.

Construction

Assemble the circuit on a general-purpose PCB. To ensure stability and prevent erratic behavior, place the timing components R4 and C4 as close as possible to Pins 6 and 7 of the NE555; this minimises stray capacitance. For a cleaner layout, use solid-core hookup wire to make neat point-to-point connections on the underside of the PCB. Alternatively, the circuit can be quickly prototyped on a breadboard as shown in the author's prototype (Fig. 1).

Testing

1. Power up. Connect a regulated 5V DC power supply to the input terminal (CON4).

2. Manual trigger. Momentarily press the TEST switch (S1). This action manually triggers the NE555 timer, which should instantly illuminate LED1 (Red).

3. Relay verification. Listen closely for a distinct clicking sound from the RL1 (5V SPDT) relay, confirming that its internal switch contacts have successfully actuated.

4. Timing check. After

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